Let H and K be ... Subgroup of ... G with $K \leq H$ Then Show that $G/H \cong (G/K)/(H/K)$

(2) long

Give an example to Show if G is a group and G/N has just two elements then we must have $|G| = 2|N|$.

(3) 3 marks Q

Let $\varphi : G \to H$ be a ... morphism. and let e_G and e_H be the identity of G and H respe. Then Show that $\varphi(e_G) = e_H$.

(4) 3

Let $G = H \times K$ be the direct product of

Two example of Normal Subgroup

Mth633 Solved by Pin2

11:30 24/8/2023

Rough Sheet

FALL/SPRING MID/FINAL TERM 20__ (Year) Date: __/__/____ Superintendent Signature:

Campus Code:_____ Student Signature:_____

Course Code:_____ Student ID:_____

① Difference b/w homomorphism & isomorphism

② Let F be additive group of all function mapping R into R and c be any real numbers Let $h: F \to R$ defined by $h(f) = f(c)$ for f in F. Show that h is homomorphism

③ Let G be a group. The set of all commutator's $aba^{-1}b^{-1}$ for $a, b \in G$ generates a subgroup C of G.

④ Define a complete group

⑤ Show that the center of a finite non-trivial p-group G is non-trivial

⑥ Let G be a group and N be normal subgroup of G If G is abelian then show that G/N is abeli

Note: Submit this sheet to Superintendent, before leaving the Examination Center.

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11:30 Long Questions 24/8/23

(1) Show that the homomorphism $h: G \to G$ is injective if and only if $\ker h = \{e\}$

(2) Show that the $Aut(G)$ of all homomorphisms of a group G is a group under composition of mappings and $Inn(G) \triangleleft Aut(G)$. Moreover, $G/Z(G) \simeq Inn(G)$

(3) Show that $(2n, +)$ is quotient gr of $(Z, +)$ by subgroup $nZ = \{nz/z$

(4) For a prime number p, show that every Group G of order p^2 is abelian

MTH633 : 19-8-2023

1, M is a maximal normal subgroup of G iff G/M is simple. pg # 864

2, Let G be a group. The set of all commutators $aba^{-1}b^{-1}$ for $a,b \in G$ generates a subgroup C of G. pg # 884

3, The center of a finite non-trivial p-group G is non-trivial. pg # 1084

4, Let X be a G-set. for $x_1, x_2 \in X$, Let $x_1 \sim x_2$ iff there exist $g \in G$ such that $gx_1 = x_2$. Then $\sim$ is an equilance relation on X. pg # 983

5, prove that
aH = bH are either equal or disjoint. pg # 564

6, Give two examples of homomorphism. Give it reason.

7, If G is a finite group and G/N has just two elements, then we must have $|G| = 2|N|$. prove that. pg # 818.

8, Give an example of Cauchy's Theorem. pg # 1046

9, Ak $\{i, -i, j, -j, k, -k\}$ wala tha statement yad nhi.

By ~Queen Sahi :).

Mcq's from files and quite easy.
and subjective main just 4 questions files sy aye.

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commutator subgroup

2) Let H be a subgroup of G, and $a, b \in G$, Show that aH are bH are either equal or disjoint.

3) Give an example to show that if G is a finite group and G/N has two elements then we must have $|G| = 2(N)$.

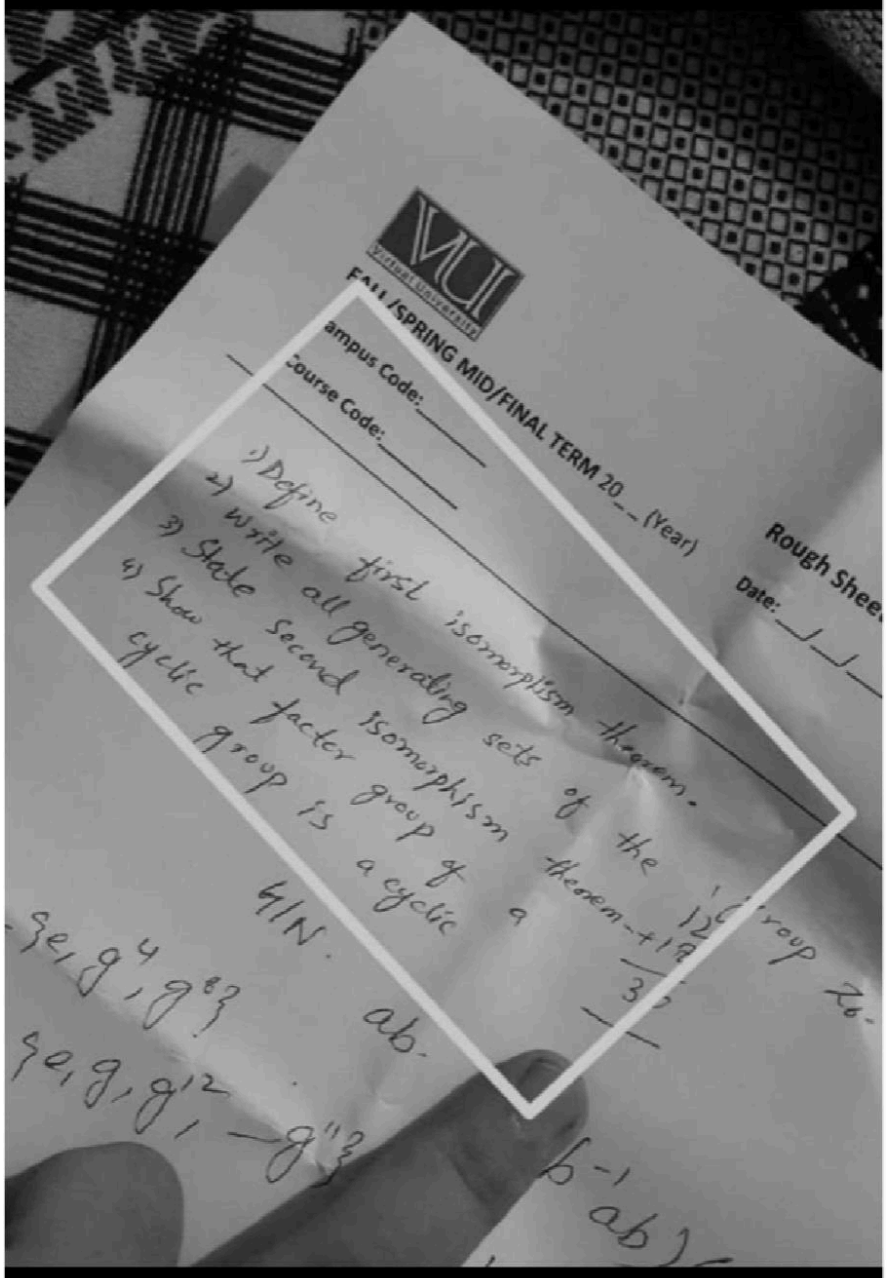
4) If N is a normal subgroup of G and if H is any subgroup of G, then Show that their join

$$H \vee N = HN = NH$$

5) Find the right cosets of subgroup of the subgroup $H = \{e, g^4, g^8\}$ in the cyclic group $G_{12} = \{e, g, g^2, \ldots g^{11}\}$

5:42 PM

You
August 21, 7:09 PM



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Date: 24.05.2023 MTH 633

(1) M is maximal normal subgroup of G if G/M is simple.

(2) Second isomorphism theorem

3. $\{e, g^4, g^8\}, \{e, g^3, g, \ldots, g^{11}\}$ $\mathbb{Z}_{12}/\mathbb{Z}_{12}$

(4) If $N \leq H \leq NH$

(5) The application of Homomorphism

(5) Give example of Lagrange Theorem

6) $(\mathbb{Z}_4, 12)$, (Value $\bar{2}$)

7) $Z(G)$ is normal and abelian subgroup of G

Action set wale